

Series Solutions of Differential Equations

Prepared by Staples Education

Practice Set: Problems and Complete Solutions

Part II — Review Guide: Practice Problems

Work the following ten problems in order. They climb through Taylor coefficients and the standard library, the radius of convergence, a power-series solution at an ordinary point and the re-indexing that produces its recurrence, the Frobenius classification and indicial equation, and finally the classical special functions — Legendre, Bessel, and the recurrence-built Hermite and Chebyshev polynomials. Solutions follow in Part III.

1. **(Maclaurin series)** Write the Maclaurin series of e^x and state the coefficient of x^4 .
2. **(Standard library)** Write the Maclaurin series of $\cos x$ in summation form.
3. **(Radius of convergence)** Use the ratio test to find the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{2^n}$.
4. **(Series at an ordinary point)** Substitute $y = \sum a_n x^n$ into $y'' + y = 0$, derive the recurrence for a_{n+2} , and identify the two independent solutions.
5. **(Re-indexing)** For $y = \sum_{n=0}^{\infty} a_n x^n$, write y'' re-indexed so every term carries the common power x^n .
6. **(Frobenius classification)** For $x^2 y'' + xy' - y = 0$, show $x = 0$ is a regular singular point, compute p_0 and q_0 , and solve the indicial equation.
7. **(Indicial roots)** Given $p_0 = \frac{3}{2}$ and $q_0 = -\frac{1}{2}$, solve the indicial equation $r(r-1) + p_0 r + q_0 = 0$.
8. **(Legendre polynomial)** For $n = 2$, Legendre's equation has a terminating solution. Find it and normalize so $P_2(1) = 1$.
9. **(Bessel's equation)** State Bessel's equation of order ν , classify the point $x = 0$, and give the indicial roots.
10. **(Classical families, capstone)** Using the Hermite recurrence $H_{n+1} = 2xH_n - 2nH_{n-1}$ and the Chebyshev recurrence $T_{n+1} = 2xT_n - T_{n-1}$ (with $H_0 = 1, H_1 = 2x, T_0 = 1, T_1 = x$), compute H_2 and T_2 , and contrast them with P_2 .

Part III — Complete Solutions

Solution 1. Every derivative of e^x equals e^x , which is 1 at $x = 0$, so $a_n = f^{(n)}(0)/n! = 1/n!$:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

The coefficient of x^4 is $\frac{1}{4!} = \frac{1}{24}$.

Solution 2. Only even powers survive, with alternating sign:

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Solution 3. With $a_n = 1/2^n$, the ratio test gives

$$\frac{1}{R} = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{1/2^{n+1}}{1/2^n} = \frac{1}{2} \implies R = 2.$$

Solution 4. Re-index $y'' = \sum_{n \geq 0} (n+2)(n+1)a_{n+2}x^n$ and add y :

$$\sum_{n=0}^{\infty} [(n+2)(n+1)a_{n+2} + a_n]x^n = 0 \implies a_{n+2} = \frac{-a_n}{(n+2)(n+1)}.$$

The a_0 branch builds $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \cos x$ and the a_1 branch builds $\sin x$, so

$$y = a_0 \cos x + a_1 \sin x.$$

Solution 5. Differentiate twice and shift the index by two so the power is x^n :

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

Forgetting the $(n+2)(n+1)$ factor here is the most common source of a wrong recurrence.

Solution 6. In standard form $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 0$, so $p = \frac{1}{x}$, $q = -\frac{1}{x^2}$. Since $xp = 1$ and $x^2q = -1$ are analytic at 0, the point is a regular singular point. Then

$$p_0 = \lim_{x \rightarrow 0} x p = 1, \quad q_0 = \lim_{x \rightarrow 0} x^2 q = -1,$$

$$r(r-1) + r - 1 = r^2 - 1 = 0 \implies r = \pm 1.$$

Solution 7. Substitute into the indicial equation and clear the fraction:

$$r(r-1) + \frac{3}{2}r - \frac{1}{2} = r^2 + \frac{1}{2}r - \frac{1}{2} = 0 \implies 2r^2 + r - 1 = 0.$$

Factor $(2r-1)(r+1) = 0$:

$$r = \frac{1}{2} \quad \text{and} \quad r = -1.$$

Solution 8. For $n = 2$ the even series terminates: $a_2 = -\frac{n(n+1)}{2}a_0 = -3a_0$ and $a_4 = 0$, so $y = a_0(1 - 3x^2)$. Normalize with $P_2(1) = 1$:

$$a_0(1 - 3) = -2a_0 = 1 \implies a_0 = -\frac{1}{2},$$

$$P_2(x) = -\frac{1}{2}(1 - 3x^2) = \frac{1}{2}(3x^2 - 1).$$

Solution 9. Bessel's equation of order ν is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

At $x = 0$ the coefficients blow up but $x^2 p$ and $x^2 q$ are analytic, so it is a regular singular point; the indicial equation $r^2 - \nu^2 = 0$ gives

$$r = \nu \quad \text{and} \quad r = -\nu.$$

Solution 10. Apply each recurrence once:

$$H_2 = 2xH_1 - 2(1)H_0 = 2x(2x) - 2 = 4x^2 - 2,$$

$$T_2 = 2xT_1 - T_0 = 2x(x) - 1 = 2x^2 - 1.$$

All three degree-2 members look alike but come from different equations and weights:

$$H_2 = 4x^2 - 2, \quad T_2 = 2x^2 - 1, \quad P_2 = \frac{1}{2}(3x^2 - 1).$$

Unit Key Takeaway

Every problem above runs the same engine: **assume an analytic (or Frobenius) ansatz, substitute, and match powers** to extract a recurrence. The ordinary-point case gives the two free constants a_0, a_1 ; the regular-singular case needs the x^r factor and its indicial roots; and the named physical equations terminate or specialize into the **classical orthogonal families**.